# THALES AZURE GENAI/ML RESOURCE KIT

## PROTECTING SENSITIVE DATA

THALES

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DRAFT

# 1 | Introduction

Generative AI's widespread appeal stems from its potential to transform industries. Businesses aim to leverage it for better customer experiences, streamlined operations, and faster processes. However, the success of these AI/ML applications hinges on data and its context, making the understanding and protection of sensitive enterprise data essential for proper and secure deployment.

Tuning or extending generative AI for specific business uses necessitates data, similar to other AI/ML workloads. A major concern for organizations is minimizing the risk of using their own data, which may contain sensitive personal information (PI/PII), for training. This personal data is often embedded within necessary contextual information for the model to function effectively.

From a security perspective there are many potential security risks associated with the implementation of GenAi systems such as Data Poisoning Definition, Backdoor insertion Definition ,Model theft definition and Prompt injection definition attacks. ***The focus of this document will be on implementing Thales solutions with AZURE capabilities to ensure sensitive data such as PII, PCI, etc will be protected.***

The Azure resource kit now includes both non-Purview local processing samples and Microsoft Purview-driven workflows. For unstructured content, Azure AI Language PII detection is used to identify the exact sensitive values in file content before Thales protection is applied. For governed Azure Storage and Azure SQL scenarios, Microsoft Purview is used to identify which assets, tables, and columns should be processed, creating a stronger link between discovery, classification, and protection.

Azure OpenAI Service provides access to OpenAI's powerful language models (like GPT-4o, GPT-3.5 Turbo) through REST APIs, SDKs, and the Azure AI Foundry portal. It allows you to integrate these models into your applications for various tasks, including:

- Content Generation: Creating articles, blog posts, marketing copy, etc.
- Summarization: Condensing large amounts of text into key points.
- Chatbots: Building conversational AI interfaces.
- Code Generation: Assisting developers with writing code.
- Translation: Converting text between languages.
- Image Understanding (with multimodal models): Analyzing and responding to image content.

Key benefits include the enterprise-grade security, scalability, and reliability of Microsoft Azure, along with responsible AI content filtering

Listed below is a diagram highlighting the Azure GenAI tooling.


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This is the overall workflow process when using the  AI Foundry:

The focus of this document will be on Azure AI Foundry which is a unified platform designed for building, deploying, and managing AI applications at scale. It brings together various Azure AI services, including Azure OpenAI Service, along with tools for:

- **Model Exploration:** Discovering and evaluating a wide range of foundation, open, task-specific, and industry models from different providers (including OpenAI).
- **Customization:** Fine-tuning models, prompt engineering, and Retrieval Augmented Generation (RAG).
- **Agent Building:** Creating AI agents with reasoning and autonomous task execution capabilities.
- **Deployment and Management:** Streamlining the deployment process and offering tools for monitoring and optimization.
- **Collaboration:** Providing a collaborative environment for AI development teams.
- **Responsible AI:** Incorporating safety filters and evaluation metrics.


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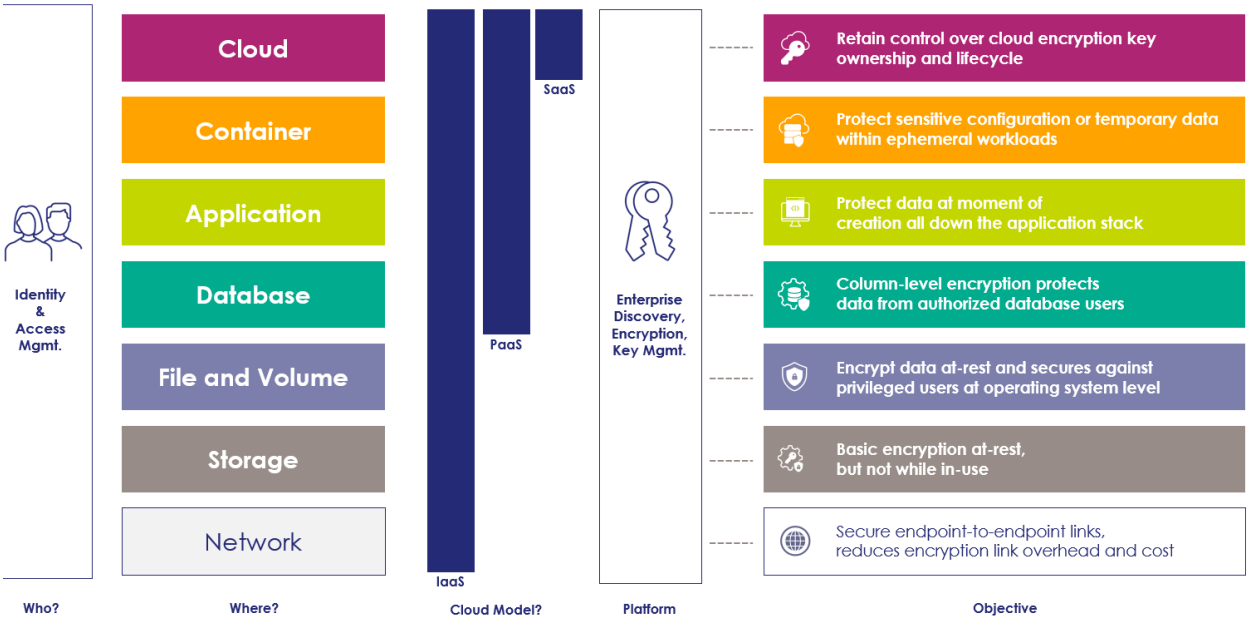
During the training phases there are situations when the data needs to be in an Azure Blob storage to be processed. Thales provides customers the ability to populate the Azure Key Vault with key material created in an external key manager (CipherTrust Manager) using customer managed keys to achieve better control and compliance. See Appendix on Bring Your Own Encryption & Hold Your Own Key for more information.

One thing to keep in mind is that using a strategy of Bring Your Own Key (BYOK) or Hold Your Own Key (HYOK) may be sufficient for many use cases, depending on the classification of your data, it may not be appropriate for use cases where the data is highly sensitive. When implementing AZURE native encryption, it is important to know what threats you are protected from with this kind of strategy. The diagram below highlights the different type of encryption and the threat you are protected from. AZURE "Native" encryption is somewhat similar to Full Disk Encryption (FDE) that customers deploy for their on-premises disk. The difference with Azure Key Vault is that instead of having one key for the entire disk AZURE provides customers the ability to have their own key for just their "service or portion of the disk" (whatever the key is protecting in the CSP). The other difference is that customers have the ability to disable the key to prevent access to the service which makes the entire service unavailable. It is important to keep in mind ***you are NOT protected once the disk is spinning and a bad actor logs onto the data center to infiltrate your systems and data***.

As seen from the diagram above when using the Azure AI Foundry capabilities some of the locations of data reside on Azure Blobs in the customer's control. Looking at the encryption stack below it is necessary to deploy a different kind of encryption, bring your own encryption (BYOE) to prevent this kind of threat. For more information on how this can be accomplished see the section titled Bring Your Own Encryption in the Appendix.

## Data Protection Use Cases




The Generative AI workflow can be broadly categorized into three distinct phases. First, the **sourcing phase** involves critical decisions regarding data collection, encompassing the gathering



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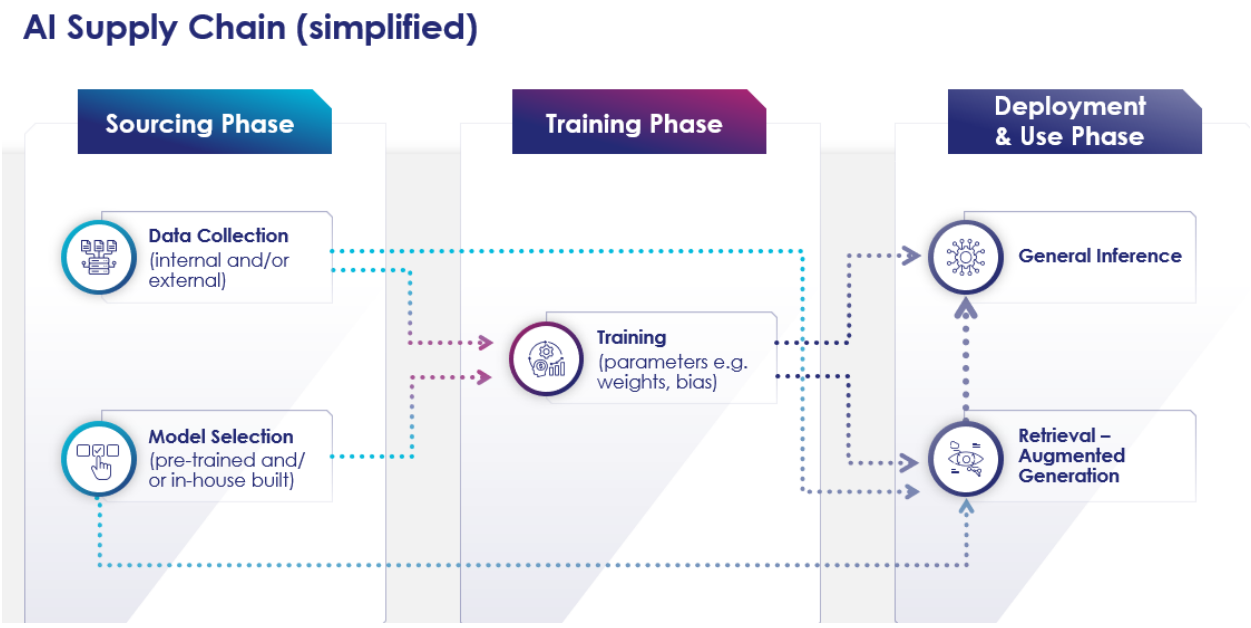
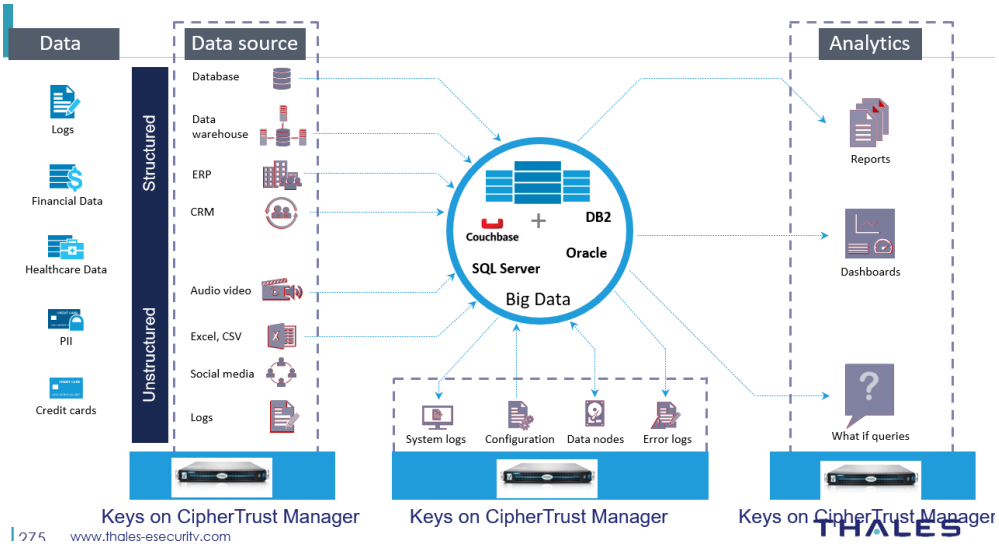
and preparation of relevant datasets, and model selection, where the appropriate pre-trained or custom model is chosen based on the desired application. Next, the **training phase** focuses on fine-tuning the selected model using the sourced data, optimizing its parameters to generate desired outputs. Finally, the **deployment phase** entails making the trained model accessible for practical use, encompassing both general inference, where the model generates outputs based on user prompts, and Retrieval Augmented Generation (RAG), which enhances responses by grounding them in external knowledge sources for improved accuracy and contextuality.



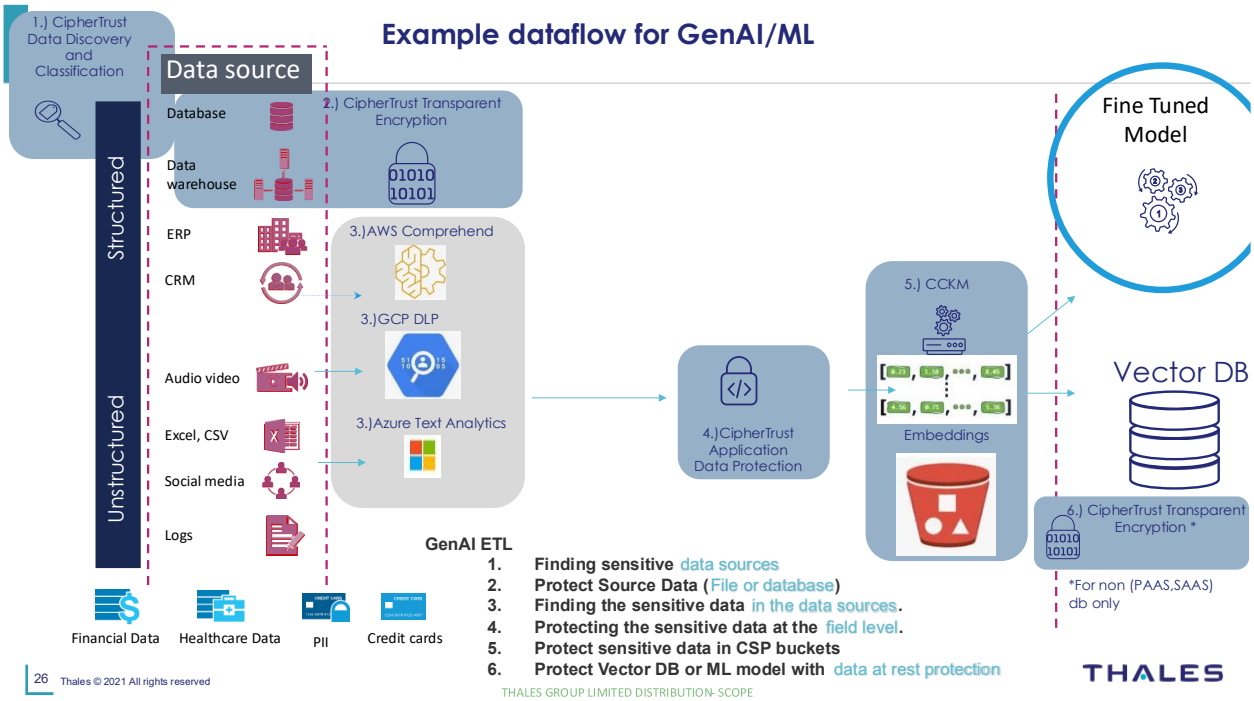

## 2 | Sourcing Phase.

The first phase is dedicated to sourcing, we want to define which data we will use for the training phase. Data appropriate for GenAI can come from several locations both structured and unstructured***. It is important to protect this data both at a file level and eventually the field or column level as well***. It is estimated that about 80% of organization data is in unstructured data sources which is something to be aware of as finding sensitive data can be a cumbersome and time-consuming process. Organizations should plan on using appropriate data discover and classification tools to identify where the sensitive data is and then once found protect it. Both Thales and AZURE provide this capability which will be explained later in this document. See Appendix section "**Finding sensitive data sources and protecting it"** for more information on this topic.

This workflow is very similar to what customers are already implementing to build data warehouses and data marts. The Extract Transform and Load (ETL) process is to extract data from source systems, consolidate, clean and protect the data in a manner that can provide value to the business. The diagram below shows a traditional ETL environment.




As can be seen above there are many locations during this process where sensitive data may reside. Depending on the classification of the data much of this information should be protected with a strategy like (BYOE) encryption that is stronger than full disk encryption (FDE) as it is **vulnerable to anyone who has access to the operating system.** There are two non-mutually exclusive strategies to protect sensitive data during this process using BYOE protecting the entire file or database and or protecting the column in a database or field in a file. Please see BYOE section for more information. The same applies for the GenAI ETL process which is shown below along with the areas where Thales can provide additional levels of protection.




As noted earlier with GenAI and ML environments it is estimated that about 80% of the data to be used is unstructured which typically has **not** been verified to be PII, PCI etc safe. The diagram above shows how the GenAi ETL process has some unique steps that are necessary to prepare the

data for training or the RAG but conceptually the process to protect the data is the same except for the fact that a higher % of the sources will be unstructured data.

1. **Finding sensitive data sources**
2. **Protect Source Data (File or database)**
3. **Finding the sensitive data in the data sources.**
4. **Protecting the sensitive data at the field level.**
5. **Protect sensitive data in CSP buckets**

## Microsoft Purview-Driven Sourcing Workflow

Microsoft Purview now fits naturally into the sourcing phase as the governance and discovery layer for Azure data sources. Purview scans supported sources such as Azure Blob Storage, ADLS Gen2, and Azure SQL, classifies sensitive assets, and gives organizations a practical way to target the right content before it is loaded into training, retrieval, or analytics workflows.

For unstructured content, Purview is used to determine which files or storage assets are sensitive enough to require additional processing. Once those assets are identified, Azure AI Language PII detection is used to locate the exact sensitive values within the file content so that Thales protection can be applied before the content is used for embeddings, vector databases, or fine tuning.

For structured data sources such as Azure SQL, Purview can identify the sensitive columns directly. This creates a second path in the resource kit where structured data can be processed in bulk using Thales protection controls without relying on per-record text scanning in the same way as the unstructured file path.

## Structured vs Unstructured Protection Paths

A practical implementation of this pattern is the non-Purview file-to-file batch flow included in the resource kit. In that sample, local file content is scanned with Azure AI Language PII detection, sensitive values are replaced with Thales-protected values, and the protected output becomes the approved source for loading a vector database or preparing a fine-tuning dataset.

This distinction is important for customers building AI pipelines. Unstructured data often needs exact in-content detection before protection, while structured data often benefits from a bulk transformation approach that protects the right fields at scale before the data is loaded into downstream services.

6. **Protect Vector DB or ML model with data at rest protection**

A common example of unstructured data is pdf file that needs to be loaded into a fine tunned model or a vector database.  As you can see below the first step is to extract the text from the document. Once this is done text scanning for sensitive data should be incorporated to the workflow converting any sensitive data to encrypted format and written back out in its original sequence replacing the sensitive **field** with the encrypted value.  Once this has been accomplished then the data will be ready for output either for training the model or for the Vectordb to be used for the RAG.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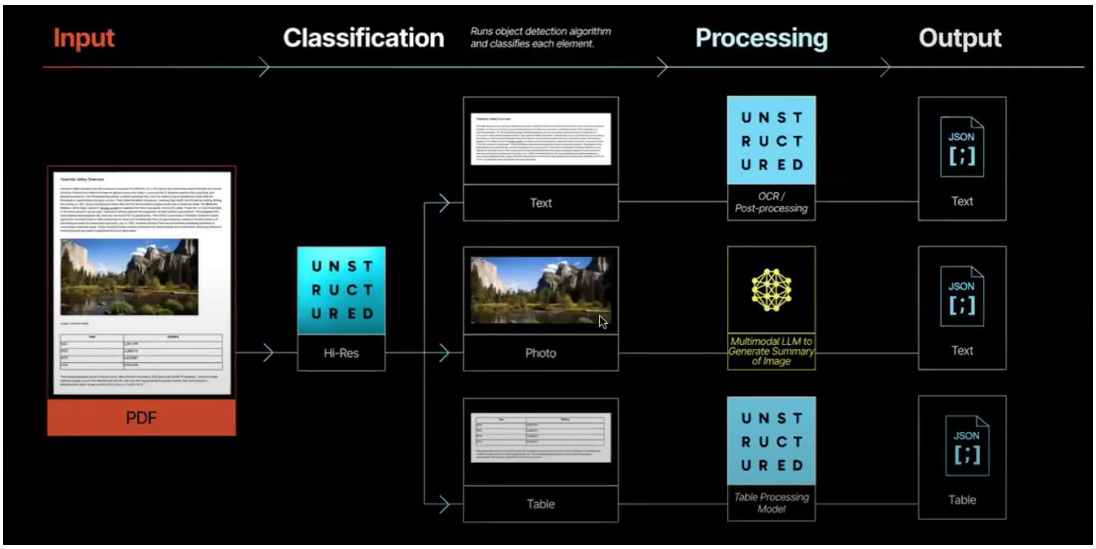
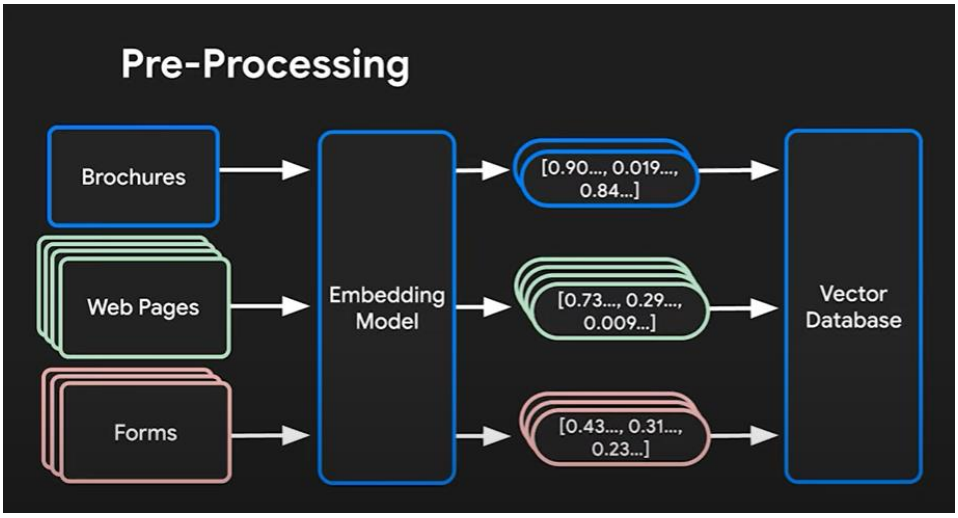



If the original sensitive data has not been protected the resulted model will inherit the same data classification as the original data set that was used to train/fine-tune it. This model can now be put under the various sophisticated jail-break, prompt injection attacks and can be forced to reveal the sensitive data it contains. Once the data has been protected the next step is to create embeddings which is done in the training phase and for prompts. The diagram below shows the overall process from sourcing the content to creating embeddings to be loaded into a vectordb and to fine tune a model.




There are various methos (encryption, fine grain authorization) to protect the sensitive data outside of the GenAI model. For example, on how this can be accomplished see Appendix "**Protecting sensitive fields in in file"**

# 3 | Training Phase

Then we have the second phase, the training phase, which is optional depending on your use case. AZURE defines the training process in 4 steps. Steps 1 and 2 have already been done in the prior phase. As you can see below step 2 is uploading the data into cloud storage, which should be



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protected at a field level before the Azure AI Language PII detection workflow and embedding process are applied to the content. The bucket can also be protected with keys in the Azure Vault and sourced by the external key manager. See sections later in this document on BYOE, BYOK and HYOK.

The training phase of the Generative AI (GenAI) process is where the model learns to understand and generate new data that resembles the data it was trained on. During step 1 the data should be formatted in a specific json format. There are a couple of formats that can be used depending on the use case. See this link for more details.

https://learn.microsoft.com/en-us/azure/ai-services/openai/how-to/fine-tuning?context=%2Fazure%2Fai-studio%2Fcontext%2Fcontext&tabs=azure-openai&pivots=programming-language-studio

This format is fine for basic evaluation or prompt-response datasets. It is commonly used for LLM fine-tuning or simple response evaluation.

```
[
  {
    "query": "What is the capital of France?",
    "response": "The capital of France is Paris."
  }
]
```

As you can see this format contains more content.

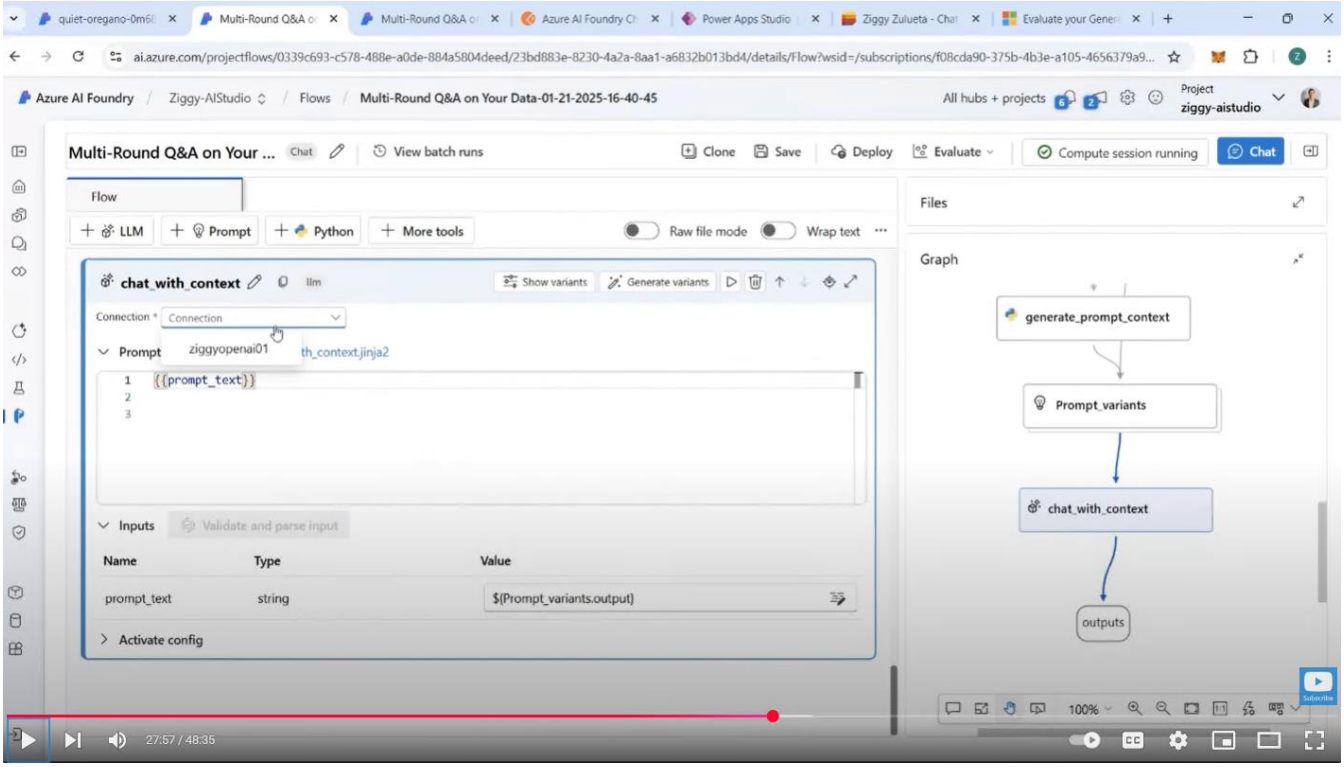
```
[
  {
    "query": "What is the capital of France?",
    "response": "The capital of France is Paris.",
    "context": "France is a country in Europe. Its capital city is Paris.",
    "ground_truth": "Paris"
  }
]
```

- **context**: Useful when evaluating **RAG (Retrieval-Augmented Generation)** systems or testing whether the model can **reason using specific background content.**
- **ground_truth**: Used for **evaluation/benchmarking.** You compare the response to the ground_truth to compute metrics like **exact match, BLEU**, or **F1 score.**
- This structure is essential for **evaluation pipelines**, such as using Azure AI Foundry's **EvalML** or **AzureML evaluations** for precision testing.

**So, which should you use?**

| Use Case | Format |
|---|---|
| Basic LLM prompt/response logging or fine-tuning | query + response |
| Evaluation with known correct answers | Add ground_truth |
| RAG testing or context-aware QA | Add context |
| Full benchmark testing (e.g., with EvalML, LangChain Eval, etc.) | All four fields |

AZURE AI Foundry has a workflow product that automates the process of training a model. Listed below is an example of the workflow.




One of the steps for the workflow is to define what bucket contains the data for training. As you can see from the screen below fine tuning a model requires uploading data to a bucket which by default includes AZURE native encryption of the data in the Azure Blob Storage which again may not be enough protection depending on your data sensitivity.

This bucket can be protected with two different strategies using the Azure Key Vault or BYOE. The process to protect the sensitive data is the same as described in the sourcing phase. Please see Appendix BYOE & Bring Your Own Keys and Hold Your Own Keys section for more information on how to deepen your security for AZURE blob storage.

Weather you decide to protect the data at a field level or not with the process above you also may consider strong levels of protection for the data stored in the AZURE buckets as there may be other data you do not want revealed to unintended audiences.

# 4 | Deployment & Use Phase.

And the last phase is the inference phase, where the decision makers will select a General Machine learning/General Inference technique or a RAG technique.

Retrieval Augmented Generation (RAG) addresses the limitations of Large Language Models (LLMs) by providing them with access to relevant, up-to-date, and domain-specific data. By grounding LLM responses in proprietary, private, or dynamic information, RAG significantly enhances accuracy and contextuality, overcoming the inherent static nature and knowledge gaps of LLMs. This is crucial for business applications requiring precise, context-aware outputs.



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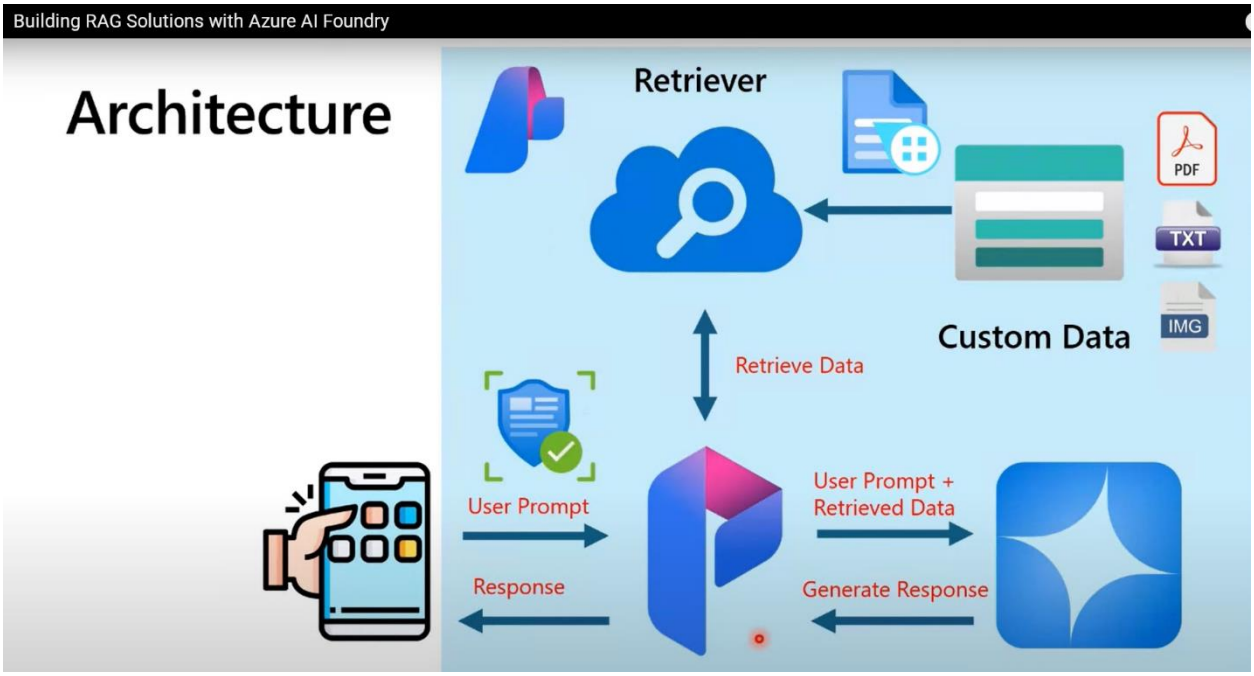
RAG also enables InfoSec team to apply the necessary security tools which are technically not possible within the model itself.

Azure AI Foundry  provides powerful APIs, SDKs, and services like Azure TextAnalytics RAG Engine and Azure TextAnalytics Pipelines that enable you to build your own automated data ingestion workflows tailored to your specific needs and data sources. These tools offer flexibility and control over the entire process.  The diagram below shows the entire RAG process.




In the diagram below the retriever is the Vector database.  The Vector database can be populated in a few ways including using the AI Foundry UI.  By default the Azure AI Search Vector DB is used.



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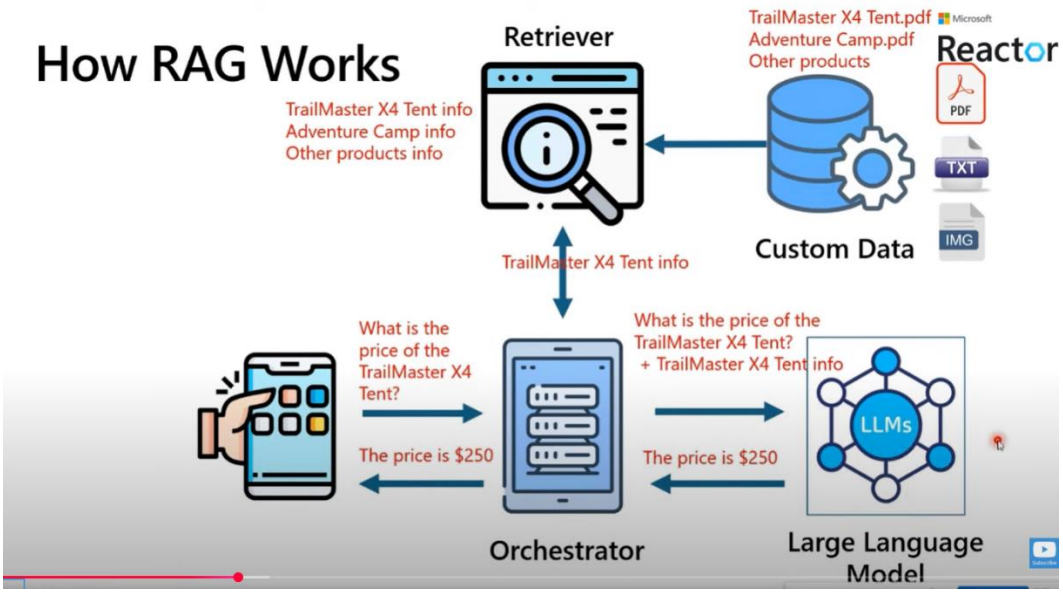


Here is a listing of all the Vector Database options from Microsoft.

-

- Store your embeddings and perform vector (similarity) search using your choice of Azure service:
  - Azure AI Search
  - Azure SQL Database
  - Azure Cosmos DB for MongoDB vCore
  - Azure SQL Database
  - Azure Cosmos DB for NoSQL
  - Azure Cosmos DB for PostgreSQL
  - Azure Cache for Redis

The Orchestrator is the prompt flow capability that AI Foundry uses to access the Vector database for relevant content and then add that to the original prompt to be sent to the LLM. Here is screenshot showing the different stages that need to take place to load the data into the Vector DB.



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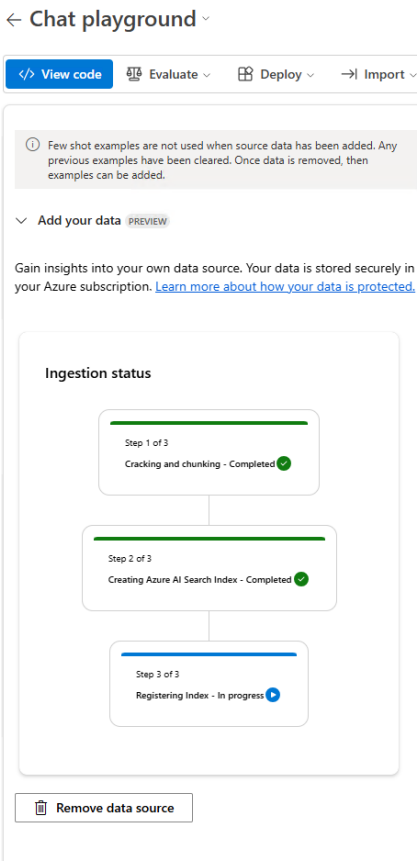
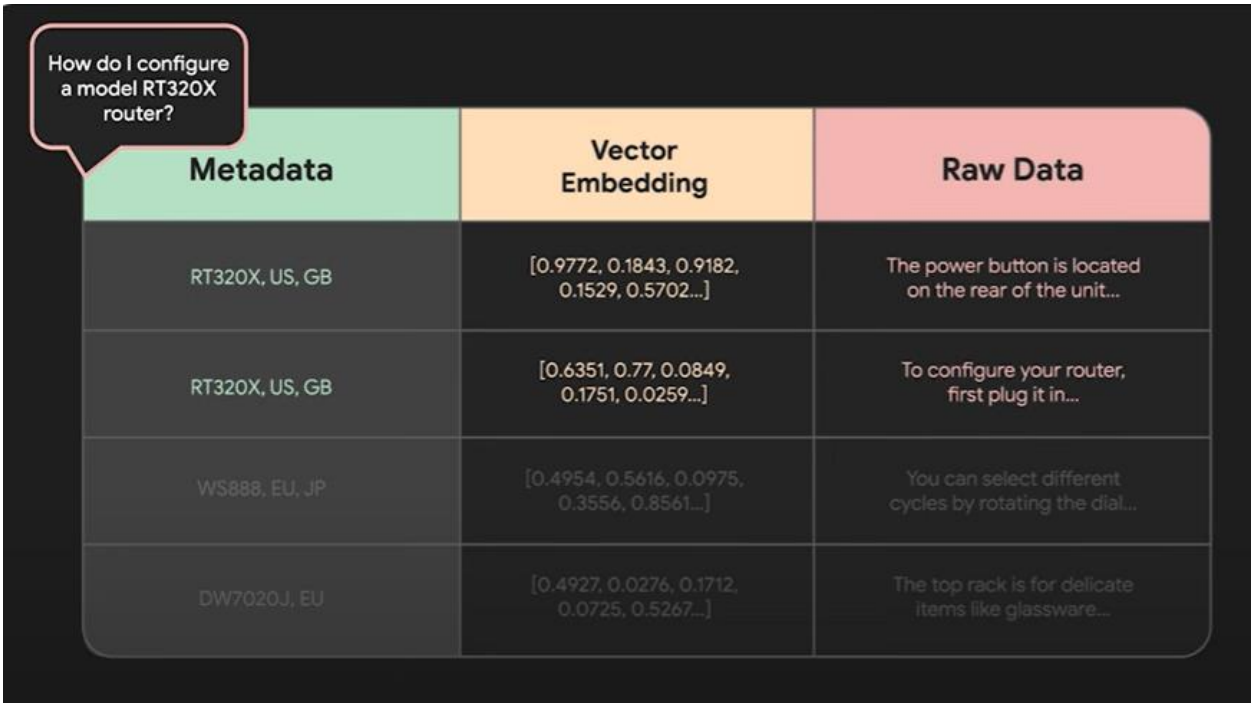



## Sensitive Data in RAG Metadata

When creating your embeddings it is possible to store custom metadata along with the metadata. You can store metadata such as author, team, department etc. to meet your enterprise prompt management needs.  If some of this information is sensitive, it may need to be encrypted to protect it from unintended usage.




The primary Azure AI Foundry  product within Azure TextAnalytics that populates and serves as the vector database for Azure AI Foundry applications is **Azure AI Search**

Azure TextAnalytics Vector Search: This is the managed vector database service in AZURE  that allows you to store and efficiently search through vector embeddings. You populate it with the vector representations of your data (text, images, audio, etc.) generated using embedding models (often also hosted on Vertex AI, like textembedding-gecko).

Azure TextAnalytics Vector Search is the dedicated, managed service within the Azure TextAnalytics ecosystem optimized for this purpose in GenAI applications. It offers features specifically tailored for vector search workloads, including efficient indexing and querying algorithms.

## Prompt Management/Engineering

Effective prompt design for language models isn't governed by strict rules, but rather by strategic techniques that can influence the model's output. However, thorough testing and evaluation are essential for achieving optimal results.

Prompt engineering is about *how* to write good prompts and prompt management is about *how* to organize and deploy those prompts effectively in both use cases sensitive data could be contained from the original data entered by the user or data that has been obtained from the Vector database or SQL queries for user specific content. Listed below are a couple of links that provide more information.
https://learn.microsoft.com/en-us/azure/ai-services/openai/concepts/prompt-engineering?context=%2Fazure%2Fai-foundry%2Fcontext%2Fcontext&tabs=chat

Microsoft uses something called “Prompt Flow” to manage the prompting in AI Foundry. Prompt flow is a development tool designed to streamline the entire development cycle of AI applications powered by Large Language Models (LLMs). Prompt flow provides a comprehensive solution that simplifies the process of prototyping, experimenting, iterating, and deploying your AI applications.

Prompt flow is available independently as an open-source project on GitHub, with its own SDK and VS Code extension. Prompt flow is also available and recommended to use as a feature within both Azure AI Foundry and Azure Machine Learning studio.

Prompt flow offers a well-defined process that facilitates the seamless development of AI applications. By using it, you can effectively progress through the stages of developing, testing, tuning, and deploying flows, ultimately resulting in the creation of fully fledged AI applications

The lifecycle consists of the following stages:

- Initialization: Identify the business use case, collect sample data, learn to build a basic prompt, and develop a flow that extends its capabilities.
- Experimentation: Run the flow against sample data, evaluate the prompt's performance, and iterate on the flow if necessary. Continuously experiment until satisfied with the results.
- Evaluation and refinement: Assess the flow's performance by running it against a larger dataset, evaluate the prompt's effectiveness, and refine as needed. Proceed to the next stage if the results meet the desired criteria.
- Production: Optimize the flow for efficiency and effectiveness, deploy it, monitor performance in a production environment, and gather usage data and feedback. Use this information to improve the flow and contribute to earlier stages for further iterations.

The process of prompt engineering is iterative and relies on testing to improve model performance. When crafting prompts, clearly define goals and anticipated results, and systematically test different approaches to identify areas for optimization.
A prompt's effectiveness hinges on two key aspects: its content and its structure. See link below for more information:
https://learn.microsoft.com/en-us/azure/ai-foundry/concepts/prompt-flow

**Here is an example of how a possible prompt interaction would work with a GenAI agent.**

**Input**:

*[Agent] Hi, my name is **Jason**, can I have your name?*

*[Customer] My name is **Valeria***

*[Agent] In case we need to contact you, what is your email address?*

*[Customer] My email is **v.racer@example.org***

*[Agent] Thank you. How can I help you?*

*[Customer] I’m having a problem with my bill.*

**De-identified Output**:

*[Agent] Hi, my name is **g4xQF**, can I have your name?*

*[Customer] My name is **3aVls3k***

*[Agent] In case we need to contact you, what is your email address?*

*[Customer] My email is **E.zu9yn@XwgFwY3.TO7***

*[Agent] Thank you. How can I help you?*

*[Customer] I'm having a problem with my bill.*

See the Appendix section (**Prompt Protection**) is an example of how to use Thales encryption capabilities to protect sensitive data either going into the prompt or from the model response. There may be scenarios where you need to re-identify a person because of the type of application or the right to removal in case of compliance. This can be done with a tagging architecture. For more information see examples provided in the git repository provided in the appendix.

# 5 | Appendix.

## Bring Your Own Keys and Hold Your Own Keys

**Native Encryption Services**: Can be deceptively simple to use, but very difficult to live with…The problem with native encryption & key mgmt. is that it is UNIQUE to the provider. These services are 100% managed by the CSPs. Also, customers have limited control and visibility over their cloud-encrypted data. There is no way to know who is accessing what and why; depending on the industry you are in, the sensitivity of data and the Cloud Service Provider, you may need to complement cloud security with additional controls for compliance. The dark blue section on the lower left is a description of AWS Data Security for GenAI outlining the native KMS encryption.



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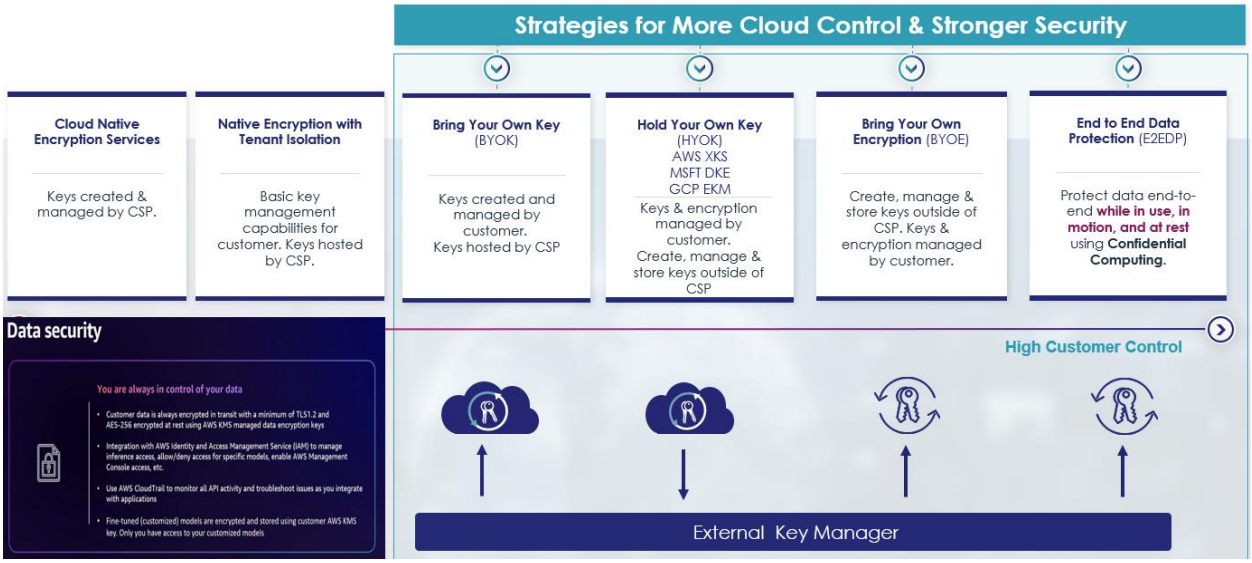
## Protection Strategies for Data in the Cloud




**Bring Your Own Key (BYOK)**: For customers that need greater control and visibility over the cloud-encrypted data, the providers offer variations of flexible key management such as Bring Your Own Key (BYOK) It is unique to the customer, depending on how they generate and manage their keys. Now, you have some control over your encrypted data because you can bring your own key material or master keys. Customers have the ability to generate and import the encryption keys or key material for their cloud-native encryption services. But still, there are gaps in visibility, control and usage of keys. Nobody know what happens after you bring your own key, if the cloud admins can access them, what happens in case of a breach or government subpoenas, etc.. Require additional tools for multi-cloud and hybrid data security & compliance.

**Hold Your Own Key (HYOK)** refers to a security model where organizations maintain full control over their encryption keys, even when using cloud services provided by a third party like **Azure AI Foundry Platform (AZURE )**, **Amazon Web Services (AWS)**, or **Microsoft Azure**. In this model, the organization uses its own key management infrastructure, often located on-premises or in a third-party environment, to generate, store, and manage encryption keys. The cloud provider does not have access to these keys, ensuring that even if the cloud infrastructure is compromised, the encrypted data remains secure and inaccessible.

## Benefits of External Key Manger

For both Bring Your Own Keys and Hold Your Own Keys you get can obtain the following benefits from the Key Manager.

1. Key Lifecycle Management
    a. Detail: Native CSP key management services has limited ability to automate the lifecycle of keys especially across multiple subscriptions
    Impact: Customers have to implement expensive manual key management processes to meet internal key security requirements

2. Attaining Compliance
    a. Detail: Insufficient authorization control or DR services to ensure keys are not accidentally or intentionally deleted
       Impact: There is too high of risk that data will be lost
    b.
3. Encryption Key Visibility
    a. Detail: Internal and external regulations require that encryption keys do not permanently reside in the cloud
       Impact: Enterprise cannot move data to the cloud where data regulations require more control of keys that are locally stored
    b.
4. Assign permissions to keys
    a. Only allow certain accounts access and operations such as encrypt,decrypt. Prevent deletes unless have a quarum. Have dates expiration dates with automatic key roation schedules, Keys have states (active,destroyed,deactivated) as approved by nist. Should be able to search on keys by attributes like algorithums, key length etc.

For more information see link below:

https://www.thalestct.com/wp-content/uploads/2022/09/Best-Practices-Cloud-Data-Protection-and-Key-Management-TCT-WP.pdf

https://cpl.thalesgroup.com/blog/cloud-security/shared-responsibility-model-cloud

# Bring Your Own Encryption (BYOE)

There are a couple of use cases to consider when using a BYOE strategy to protect sensitive data for GenAI platforms. The first is protecting the entire file or all files in a particular directory or S3 bucket and only allowing a certain process or user to access the content. The second is to protect the fields in a file that contain sensitive data. For working example on how to protect sensitive fields in a file see Appendix section on Examples.

## Protecting the entire file.

Thales provides a capability called CipherTrust Transparent Encryption(CTE) to protect the entire file in a directory or S3 bucket. See links for more information on CTE.

https://cpl.thalesgroup.com/encryption/transparent-encryption

https://www.youtube.com/watch?v=VzRxrhUZfc0

https://www.thalestct.com/wp-content/uploads/2022/09/aws-s3-with-cte-sb.pdf

https://www.thalestct.com/wp-content/uploads/2022/09/avoiding-amazon-s3-leaks-wp.pdf

## Protecting fields in a file.

Thales provides several different application encryption and tokenization capabilities.

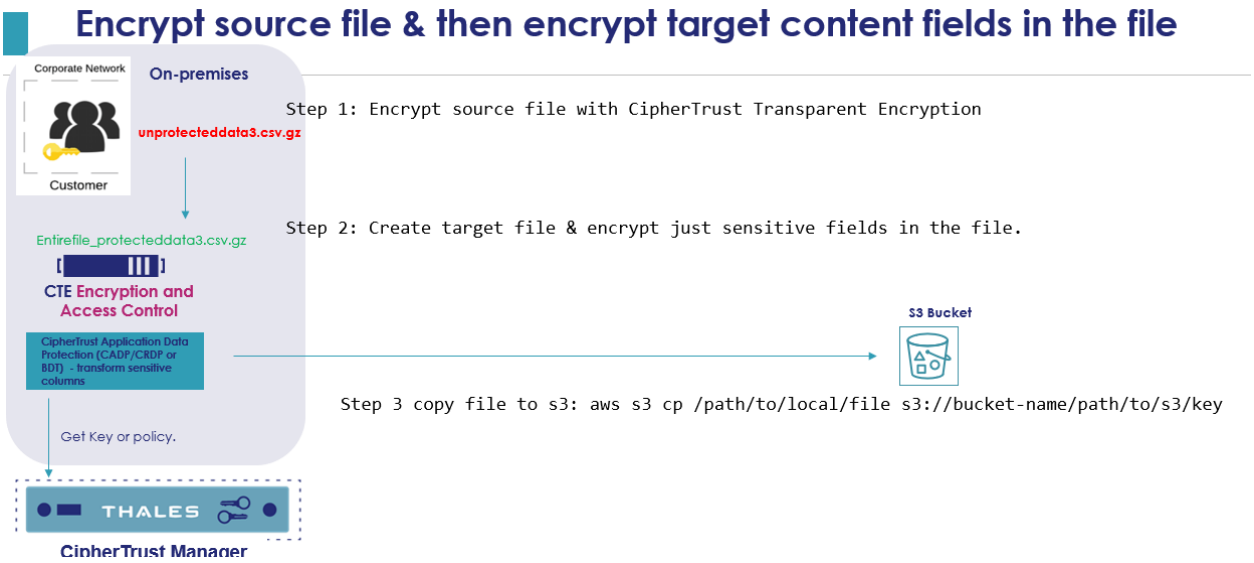
See links below for more information.

https://cpl.thalesgroup.com/encryption/ciphertrust-application-data-protection

https://cpl.thalesgroup.com/encryption/tokenization

***Note for some use cases it might be necessary to protect the file with both types of encryption depending on what stage of the process it is during the ETL process. Listed below are a couple of options on how this can be accomplished.***

### Option 1. Encryption on premises.




For working example see **Protecting sensitive fields in in file** in the Appendix Examples section.

# Finding sensitive data sources and protecting it.

This content in this section focuses on encryption of sensitive data for AI and ML environments. It will explain how data needs to be protected at a field level in a file before the content is made available to a vector database or used for training an AI model. Since the content is specifically used by the models encrypting it at a field level is the most secure and appropriate for this use case.

When protecting sensitive data for general business use protecting sensitive data in a file at a field level may or may not be appropriate since decrypting on the fly and incorporating RBAC may not be available to the applications using the files. There are other methods such as Thales CipherTrust Manger Transparent Encryption (CTE) that may be more appropriate for these kinds of use cases.



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As noted earlier it is critical that organizations find data sources that have sensitive information to avoid breaches. Microsoft Purview and Azure AI Language PII detection solve different parts of the same problem. Purview is the governance and discovery layer, while Azure AI Language PII detection is the fine-grained in-content detection API for unstructured and semi-structured text.

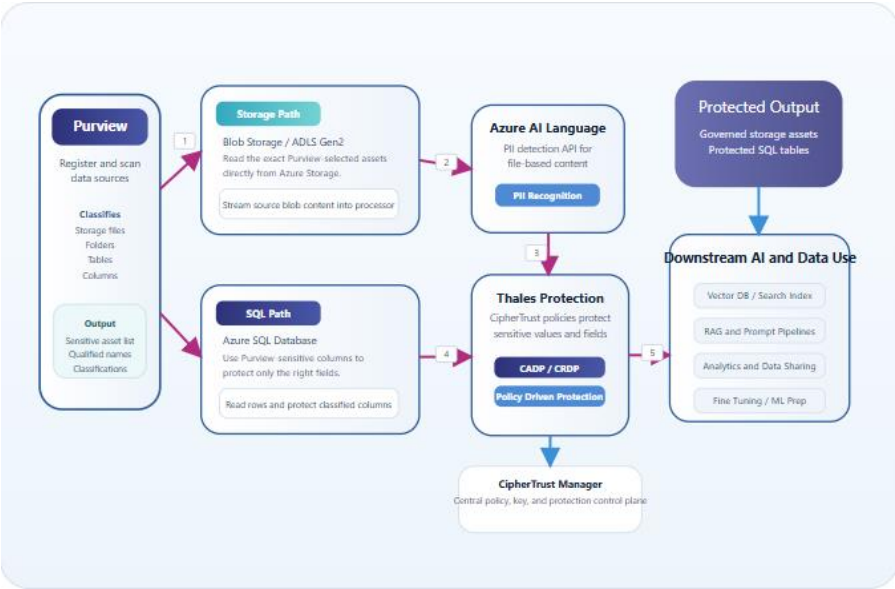
Purview is strongest when organizations need to scan supported enterprise data sources and determine which storage assets, tables, or columns are sensitive. It classifies files, folders, tables, and columns and is a very good fit for deciding what should be processed before data is moved into training, retrieval, or analytics workflows.

Azure AI Language PII detection is strongest when the exact sensitive values inside a file or text stream must be identified. In the resource kit, this is the capability used by the local file-to-file batch processor and the governed storage workflow to locate names, email addresses, phone numbers, social security numbers, addresses, dates, and similar values before Thales protection is applied.

For structured sources such as Azure SQL, Purview can identify the sensitive columns directly. This reduces the need to treat the source like free-form text and creates a more efficient path for structured bulk protection, including use cases where Thales Batch Data Transformation is the better fit for database or CSV processing.

The updated Azure resource kit therefore supports two governed patterns. The first is an unstructured path: Purview identifies sensitive storage assets, Azure AI Language PII detection identifies the exact values in the file content, and Thales protects them. The second is a structured path: Purview identifies sensitive tables or columns, and Thales protection is applied at the structured field level before downstream use.

In short, Purview tells you what to inspect, Azure AI Language PII detection tells you what to replace in file content, and Thales CipherTrust provides the protection and control model for both structured and unstructured workflows.



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# Examples

For structured database or CSV transformation use cases, the governed Purview path can also be paired with Thales Batch Data Transformation concepts. In that model, Purview identifies the sensitive columns or structured fields, and Thales bulk transformation processes protect the structured data at scale before it is loaded into analytics, vector, or model-training workflows.

## Overview of included examples

The Azure resource kit now includes examples for non-Purview local file processing, Purview-driven storage and Azure SQL processing, prompt protection for Azure OpenAI, and structured bulk transformation patterns. The examples below can be used individually or combined into a broader governed AI data-protection story.

## Example 1. Prompt protection for Azure OpenAI

The ThalesAzureOpenAIClientPromptExample shows how to protect prompt content before it is sent to Azure OpenAI and how output content can be reviewed again for sensitive values. This is useful for customer conversations where prompt management, retrieval results, and response safety must be discussed together.

## Example 2. Purview-driven Azure Storage and Azure SQL processing

The resource kit also includes the ThalesAzurePurviewBatchProcessor class. In the storage path, Microsoft Purview identifies which Blob Storage or ADLS Gen2 assets are sensitive, Azure AI Language PII detection identifies the exact sensitive values in file content, and Thales protection is applied before the content is written back to a governed target location. In the SQL path, Purview identifies the sensitive columns so that structured data can be protected more efficiently before downstream AI or analytics use.

## Example 3. Non-Purview local file-to-file processing

This example uses the ThalesAzureProtectRevealBatchProcessor class to read local files, use Azure AI Language PII detection to identify sensitive values in file content, protect those values with Thales CipherTrust, and write protected output to a local destination. This is a practical example for preparing files before vector database ingestion or model fine-tuning preparation.

## Protecting sensitive fields in in file

As mentioned earlier it is estimated about 80% of data in an organization is unstructured files such as pdf text and csv files.  It is important to convert the sensitive data in those files before the content is used in a vector database or used to fine train a model.  Here is an example of how this can be accomplished using the Thales Application Encryption along with the AZURE Text Analytics API to find sensitive data in a file.

```
https://github.com/ThalesGroup/CipherTrust_Application_Protection/blob/master/demos/genai/azure/s
rc/main/java/com/example/ThalesAzureProtectRevealBatchProcessor.java
```

## Prompt Protection

This java example uses a couple of helper classes, ThalesProtectRevealHelper to control settings for the CiphterTrust Manager and associated profile information and ContentProcessor to control settings for the content to be protected. To view the entire solution see the following github repository: ???

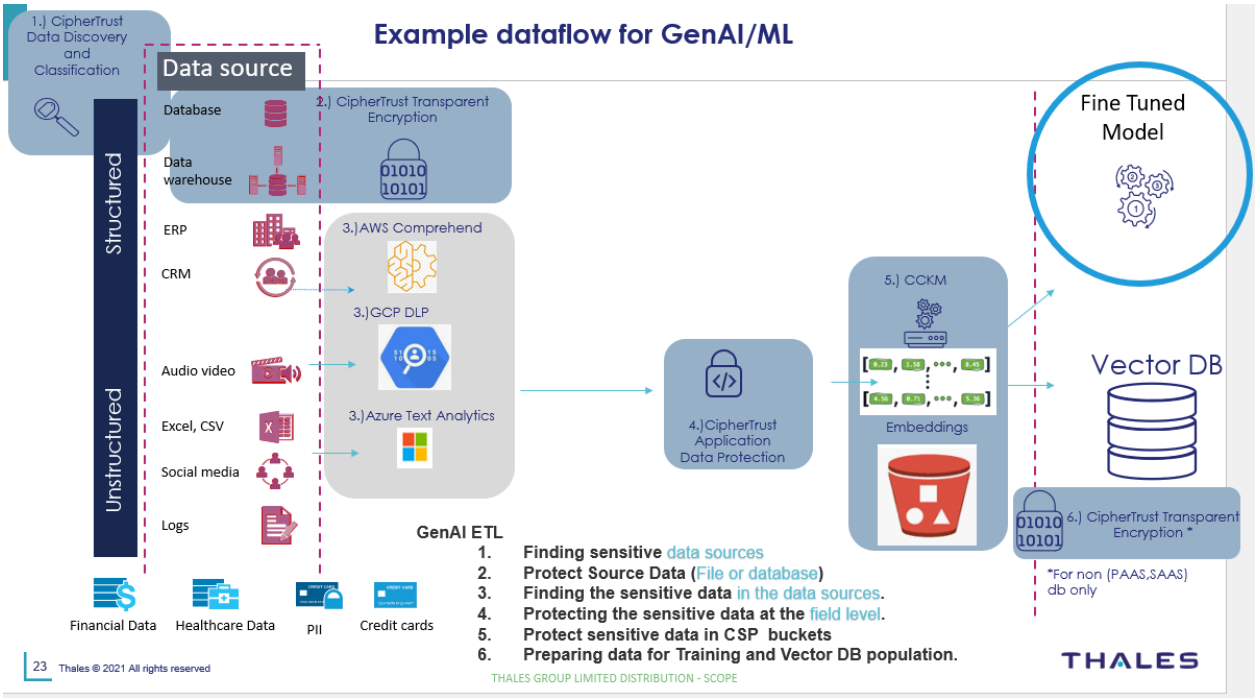
https://github.com/ThalesGroup/CipherTrust_Application_Protection/blob/master/demos/genai/azure/src/main/java/com/example/ThalesAzureOpenAIClientPromptExample.java

Output:

```
- PII Entity: Kathi Wolf, Category: Person, Confidence: 0.85
 - PII Entity: Burney Circle, Category: Address, Confidence: 0.77
 - PII Entity: OH      24114, Category: Address, Confidence: 0.66
 - PII Entity: becker.drury@hotmail.com, Category: Email, Confidence: 0.80
 - PII Entity: 7/7/1992, Category: DateTime, Confidence: 0.90
 - PII Entity: 203-04-0501, Category: USSocialSecurityNumber, Confidence: 0.65
        String inputText = Explain 'rubber duck debugging' in one line to RU5DY2hhcg==-
17:10010001Hm0W bzIA   Burney Circle  Mertzmouth     OH      24114   328.639.0623x3743
        RU5DY2hhcg==-31:10010001m4Uly.x7r1S@8639OwP.uZq      7/7/1992       5.32755E+15    15
        203-04-0501
Full Response:
{"choices":[{"content_filter_results":{"hate":{"filtered":false,"severity":"safe"},"self_harm":{"
filtered":false,"severity":"safe"},"sexual":{"filtered":false,"severity":"safe"},"violence":{"fil
tered":false,"severity":"safe"}},"finish_reason":"stop","index":0,"logprobs":null,"message":{"con
tent":"Rubber duck debugging is a method where a programmer explains their code line-by-line to a
rubber duck in order to identify
errors.","role":"assistant"}}],"created":1752079998,"id":"chatcmpl-
BrSKkSmhpRfVbpi403MpHz5wpNH5Q","model":"gpt-4-turbo-2024-04-
09","object":"chat.completion","prompt_filter_results":[{"prompt_index":0,"content_filter_results
":{}}],"system_fingerprint":"fp_5603ee5e2e","usage":{"completion_tokens":26,"prompt_tokens":118,"
total_tokens":144}}

{"created":1752079998,"prompt_filter_results":[{"content_filter_results":{},"prompt_index":0}],"u
sage":{"completion_tokens":26,"prompt_tokens":118,"total_tokens":144},"model":"gpt-4-turbo-2024-
04-09","id":"chatcmpl-
BrSKkSmhpRfVbpi403MpHz5wpNH5Q","choices":[{"content_filter_results":{"self_harm":{"severity":"saf
e","filtered":false},"hate":{"severity":"safe","filtered":false},"sexual":{"severity":"safe","fil
tered":false},"violence":{"severity":"safe","filtered":false}},"finish_reason":"stop","index":0,"
message":{"role":"assistant","content":"Rubber duck debugging is a method where a programmer
explains their code line-by-line to a rubber duck in order to identify
errors."},"logprobs":null}],"system_fingerprint":"fp_5603ee5e2e","object":"chat.completion"}

Extracted Content:
Rubber duck debugging is a method where a programmer explains their code line-by-line to a rubber
duck in order to identify errors.

Now checking for PII in output :
Rubber duck debugging is a method where a programmer explains their code line-by-line to a rubber
duck in order to identify errors.
 - PII Entity: sam@aol.com, Category: Email, Confidence: 0.80
clean output = Rubber duck debugging is a method where a programmer explains their code line-by-
line to a rubber duck in order to identify errors. RU5DY2hhcg==-18:1001000wt9@FBl.52X

Content Filter Results:
- self_harm: filtered=false, severity=safe
- hate: filtered=false, severity=safe
- sexual: filtered=false, severity=safe
- violence: filtered=false, severity=safe
```

## Overall workflow

The diagram below shows the entire process




```
Here are some samples that can be used to implement the above.

1.) Run Thales DDC or other datasoruce discovery tool to find datasources that have sensitive
data.
2.) Use Thales CTE to protect those data sources.
3.) Run Thales DDC Report to export a list of datasources with sensitive data.
4.) If any of the sources are of file types excel,word,pdf run sample code
ExtractFilesFromDDCReport.java to generate a list of the datsources that need to be converted to
text.
5.) Run sample code ConvertToTextBatchProcessor.java to convert files found in step for to text
version.
6.) Depending on customer preference for CSP (AWS,GCP,Azure) run appropriate sample code
ThalesCSPProtectRevealBatchProcessor.java to find sensitive fields in datasources and create a
new output with sensitive data protected.
7.) Now load protected data into appropriate AWS,GCP, Azure training data upload.
```

## Thales Protect/Reveal Options

The solutions provide in this document for field level protection are based on two application development offerings from Thales CRDP and CADP for Java.

### 1.)CRDP

CipherTrust RESTful Data Protection (CRDP) is a RESTful webservice that protects sensitive data using encryption, tokenization, or data generalization. CRDP is designed from the ground up to seamlessly fit with existing cloud and on-premise applications. CRDP is deployed as a container and performs a wide range of data protection functions on the sensitive data including data masking, pseudonymization, and redaction. CRDP is easy to deploy, configure, manage, and migrate.
https://thalesdocs.com/ctp/con/crdp/latest/index.html

## 2.) CipherTrust Application Data Protection for Java (CADP for Java)

Is a Java Cryptography Extension (JCE) provider that enables users to integrate the Key Manager's capabilities into their Java applications. CADP for Java is available in two variants:

- **Centrally Managed APIs**
- **Traditional APIs**

**The examples provided in this document are based on Centrally Managed APIs.**

This is our new set of APIs available from CADP for Java 8.18.1 release. CADP for Java offers simplified APIs (Protect/Reveal) to perform cryptography operations. CADP for Java integrates seamlessly with **Application Data Protection** on CipherTrust Manager, enabling organizations to centrally configure, manage, and enforce data-centric cryptographic policies in a reusable, human-readable format, providing flexibility and ease of management.
Protection policy (ciphers, keys, IV, tweak, and so on) defines how to protect the sensitive data. Whereas, Access policy determines who can view sensitive data and how (plaintext, ciphertext, custom masking format or redacted).
Centrally Managed APIs offer several advantages over traditional APIs, including:

- Developers do not need to understand cryptographic parameters (cipher, key, IV, Nonce, Tweak, and so on) to protect data as the Data Security Admins are responsible to handle configurations and policies.
- Each deployed application with CADP for Java is visible on CipherTrust Manager (providing a Single Pane of Glass view).
- Data Security Admins gain Crypto Agility, enabling real time changes to cipher, keys, and parameters.

https://thalesdocs.com/ctp/con/cadp/cadp-java/latest/admin/index.html#centrally-managed-apis

The code provided in this document provides examples using both the CRDP and CADP for Java Centralized API's. As you can see from the code there are two helper classes that make it easy to switch between either api.

```
// Initialize ThalesProtectRevealHelper. Two options are commented out, showing different
// implementations (CADP vs. REST). The REST implementation is currently active.
// ThalesProtectRevealHelper tprh = new ThalesCADPProtectRevealHelper(keymanagerhost, crdptkn,
null, policyType,showmeta);

// Using ThalesRestProtectRevealHelper for content protection and revelation.
ThalesProtectRevealHelper tprh = new ThalesRestProtectRevealHelper(crdpip, metadata,
policyType,showmeta);

Note: For each option there will be different properties required please refer to the code
samples for detailed explanations.
```

*Thales Protect/Reveal Sample Code*

*https://github.com/ThalesGroup/CipherTrust_Application_Protection/blob/master/crypto/java/ProtectRevealSample.java*

*https://github.com/ThalesGroup/CipherTrust_Application_Protection/blob/master/crypto/java/ProtectRevealBulkDataSample.java*

*Thales ReProtect or Rekey Sample Code*

*https://github.com/ThalesGroup/CipherTrust_Application_Protection/blob/master/crypto/java/ReprotectSample.java*

## copilot vs AI Foundry

Microsoft Copilot and Azure AI Foundry are both powerful AI offerings from Microsoft, but they are designed for different users and use cases, and they operate at different levels of the technology stack.[1] Here's a breakdown of the key differences:

**Summary:**

| Feature | Microsoft Copilot | Azure AI Foundry |
|---|---|---|
| **Primary Purpose** | An AI assistant to boost productivity for end-users. | A platform for developers to build and manage custom AI solutions. |
| **Target User** | General users, business professionals. | Developers, data scientists, AI professionals. |
| **User Experience** | Conversational, integrated into Microsoft 365 apps. | Code-first, with a portal and SDKs for deep customization. |
| **Level of Control** | Limited; core functionality is managed by Microsoft. | Full control over models, data, and architecture. |
| **Use Case Example** | Drafting an email, summarizing a meeting, generating a PowerPoint presentation. | Building a custom AI-powered chatbot for a specific business process, creating a predictive maintenance system, developing a multi-agent workflow. |